

Task 'Unit 4 Design Patterns II - Structural Patterns' Unit 4

Task Collaborative Discussion 1

Structural design patterns helps to improve maintainability and flexibility. The Adapter Pattern helps incompatible systems communicate, for instance, in a payment scenario, an adapter converts REST JSON requests into SOAP messages for a legacy gateway, this avoids rewriting the old system.

(Lott and Phillips, 2021)

```
```python
```

```
class LegacyPay:
```

```
 def pay_soap(self, data): print("SOAP:", data)
```

```
class Adapter:
```

```
 def __init__(self, legacy): self.legacy = legacy
```

```
 def pay_json(self, data): self.legacy.pay_soap(str(data))
```

```
Adapter(LegacyPay()).pay_json({"amount":100})
```

```
```
```

The Bridge Pattern separates abstraction from implementation, the remote control, can work with different devices without changing its code, this reduces duplication and supports scalability.

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```
```python
```

```
class TV:
```

```
 def on(self): print("TV on")
```

```
class Radio:
```

```
 def on(self): print("Radio on")
```

```
class Remote:
```

```
 def __init__(self, device): self.device = device
```

```
 def power(self): self.device.on()
```

```
Remote(TV()).power()
```

```
```
```

The Composite Pattern treats files and folders equally, for example, it simplifies recursive operations such as displaying or searching a directory tree by treating folders containing files or other folders equally.

```
```python
```

```
class File:
```

```
 def __init__(self, name): self.name=name
```

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```
def show(self): print(self.name)
```

```
class Folder:
```

```
 def __init__(self, name):
```

```
 self.name=name; self.items=[]
```

```
 def add(self, item): self.items.append(item)
```

```
 def show(self):
```

```
 print(self.name)
```

```
 for i in self.items: i.show()
```

```
docs=Folder("Docs")
```

```
docs.add(File("notes.txt"))
```

```
docs.show()
```

```
...
```

(Refactoring Guru, 2024)

Using the above techniques, help to develop a better and a more reusable design that improves readability, extensibility and code maintenance, with the Adapter enabling better integration, Bridge producing independent variations and Composite simplifying hierarchical structures.

(Lott and Phillips, 2021)

## **Task ‘Unit 4 Design Patterns II - Structural Patterns’ Unit 4**

### **References**

- Steven F. Lott, D. P. (2021) Python Object-Oriented Programming: Build robust and maintainable object-oriented Python applications and libraries, 4th Edition. Packt Publishing.
- Refactoring Guru (2024) Adapter Pattern. Available at: Refactoring Guru Adapter Pattern, available at: <https://refactoring.guru/design-patterns/adapter> [Accessed: 16 May 2026].
- Refactoring Guru (2024) Bridge Pattern. Available at: Refactoring Guru Bridge Pattern, available at: <https://refactoring.guru/design-patterns/bridge> [Accessed: 16 May 2026].
- Refactoring Guru (2024) Composite Pattern. Available at: Refactoring Guru Composite Pattern, available at: <https://refactoring.guru/design-patterns/composite> [Accessed: 16 May 2026].

### **Use of Digital Tools**

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